Halcyon Robotics Field Operations Handbook

Version 4.2, effective 1 March 2026. Owner: Field Operations Office (FOO).

Halcyon Robotics builds and operates autonomous inspection robots for utilities, rail operators and wind-farm owners. This handbook applies to every field operator, crew lead and contractor who deploys Halcyon equipment at a customer site. Where this handbook and a customer's site rules disagree, the stricter rule applies.

The company runs two robot families. The HX-7 is a quadcopter inspection drone used for power lines, towers and turbine blades. The Talon-3 is a tracked ground rover used inside substations, tunnels and rail yards where flying is not allowed. Both families report telemetry to the Halcyon fleet platform through regional telemetry gateways.

Operators are organized into regional crews. Each crew has a crew lead who signs off the daily deployment plan, and a duty manager who can be reached around the clock through the on-call rotation described in section 8.

1. Equipment overview: HX-7 and Talon-3

HX-7 inspection drone. Maximum takeoff weight 6.8 kg. The HX-7 can stay airborne for up to 38 minutes on a single battery pack in calm conditions; crews must plan missions for no more than 30 minutes to keep a reserve. Maximum payload is 1.2 kg, which covers the standard zoom camera or the thermal camera, but not both at once. Wind limit for operations is 12 m/s sustained or 15 m/s gusts. The HX-7 uses the LX-200 LiDAR module for obstacle avoidance near conductors.

Talon-3 ground rover. Operating time is about 6 hours per battery set, top speed 1.5 m/s, and it can climb grades up to 25 degrees. The rover carries the same thermal camera as the HX-7 plus an acoustic sensor used to detect partial discharge in substation equipment. It is rated IP66 and may operate in rain, but not in standing water deeper than 10 cm.

Both robots are identified by a fleet ID painted on the chassis (for example HX7-0412 or T3-0091). The fleet ID must be written on every checklist, log and report.

2. Pre-deployment checks

Before every deployment the operator completes the pre-flight checklist on form OPS-12. OPS-12 covers battery health, propeller condition, firmware version, GPS lock, compass calibration and the geofence for the site. A deployment may not start until the crew lead has countersigned OPS-12.

Firmware on every robot must be within one minor version of the current fleet release. Robots that fall further behind are grounded until updated. Compass calibration is required whenever the robot has been moved more than 50 km since its last flight.

Operators must also confirm that the customer's site permit is valid for the date and that the emergency landing zones in the deployment plan are clear.

3. Battery safety

All Halcyon robots use lithium-ion battery packs. Packs are stored between 15 and 25 degrees Celsius in the fireproof cabinets supplied with each crew vehicle. Never charge a pack that is swollen, punctured or warmer than 45 degrees Celsius; tag it red and isolate it in the sand bin.

For shipping by road or air, packs must be discharged to 40 percent state of charge or less. Packs above 40 percent may not be shipped under any circumstances, even between Halcyon sites.

A pack is retired after 300 charge cycles or when its capacity falls below 80 percent of the rated value, whichever comes first. Retired packs go back to the regional depot for recycling; they must not be thrown away with ordinary waste.

4. Maintenance schedule

Routine maintenance on the HX-7 is due every 120 flight hours and is recorded on form OPS-23, the maintenance log. Propellers are replaced every 300 flight hours or immediately if any nick or crack is found during the OPS-12 check. Motors are inspected at 600 flight hours.

The Talon-3 tracks are inspected every 200 operating hours and replaced at 1,500 hours. Gearbox oil is changed once a year.

Only technicians holding certification level L2 or higher may sign off maintenance on form OPS-23. Level L1 operators may perform the pre-flight checklist but may not open sealed components.

5. Incident and near-miss reporting

Any incident involving injury, property damage, loss of a robot, or a fly-away must be reported within 24 hours on form OPS-19, the incident report. The crew lead also phones the duty manager immediately for any injury or any loss of control near people.

Near-misses, such as a robot entering a geofence buffer or a battery warning during flight, are reported within 72 hours on the same OPS-19 form with the box "near-miss" ticked. Reports are blame-free: the goal is to fix systems, not to punish individuals.

Telemetry outages that affect more than 50 robots are handled as major incidents by the platform on-call engineer and get a written postmortem within 10 business days.

6. Data handling and customer privacy

Inspection imagery belongs to the customer. Raw video and photos from customer sites are uploaded to the customer's project space within 48 hours and deleted from robot storage and crew laptops after upload.

Any footage that unintentionally captures people or neighbouring private property is deleted after 30 days unless the customer requests it for an investigation. Footage may never be posted on social media or used in marketing without written permission from the customer.

Crew laptops use full-disk encryption. A lost laptop or robot storage card is reported as an incident on OPS-19.

7. Training and certification

Operators progress through three certification levels.

Level L1 requires 8 hours of classroom training and a written exam. L1 operators may fly the HX-7 only under direct supervision.

Level L2 requires at least 40 supervised flight hours after L1, a practical exam with a senior examiner, and a first-aid certificate. L2 operators may fly unsupervised in daylight and sign maintenance logs.

Level L3 requires 120 total flight hours, a night-operations exam and a recommendation from a crew lead. Only L3 operators may fly at night or beyond visual line of sight where the customer's permit allows it. Certifications are renewed every 24 months.

8. On-call rotation, travel and expenses

Every region keeps one duty manager on call at all times. Shifts run for one week, Monday 09:00 to the following Monday 09:00 local time. The on-call phone must be answered within 15 minutes.

Travel is booked through the company travel desk. The meal per diem for field travel is 68 US dollars per day; hotel costs are reimbursed at actual cost up to the city limit published by Finance. Expense claims are filed on form FIN-4 within 30 days of the trip.

Crew vehicles must carry the fire extinguisher, the sand bin for damaged batteries and the first-aid kit at all times.